Q1 Equilibrium

a). Solve for the equilibrium price P^* and quantity Q^* analytically.

Demand and Supply Equations

- **Demand Equation:** $Q_D = \alpha_0 + \alpha_1 P + \alpha_2 Y + U_D$
- **Supply Equation:** $Q_S = \beta_0 + \beta_1 P + \beta_2 W + U_S$

Equilibrium Condition

At equilibrium, $Q_D = Q_S$.

Substituting the demand and supply equations:

$$\alpha_0 + \alpha_1 P + \alpha_2 Y + U_D = \beta_0 + \beta_1 P + \beta_2 W + U_S$$

Solving for P^* and Q^*

Rearranging:

$$(\alpha_1 - \beta_1)P = \beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D$$

Solving for P^*:

$$P^* = \frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D}{\alpha_1 - \beta_1}$$

Solving for Q^*:

$$Q^* = \alpha_0 + \alpha_1 P^* + \alpha_2 Y + U_D = \beta_0 + \beta_1 P^* + \beta_2 W + U_S$$

b) Provide a brief interpretation of these equilibrium expressions.

These are simultaneous equations, the equilibrium is decided by both demand-and-supply-related parameters and variables – willingness to pay/sell, price of the demand/supply shifters, together with market-specific unobserved factors U_S and U_D

Equilibrium Price P^*

$$P^* = \frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D}{\alpha_1 - \beta_1}$$

Interpretation:

The numerator $\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D$ represents the net effect of differences in the minimum willingness to sell and the maximum willingness to pay $\beta_0 - \alpha_0$, demand and supply shifters $\beta_2 W - \alpha_2 Y$, and unobserved heterogeneity $U_S - U_D$.

- $\beta_0 - \alpha_0$ is always negative, a rise in the minimum willingness to sell (β_0) increases P^* as it drives down the supply and vice versa.
- if the product is a normal good, then an increase in consumers' income (Y) reduces P^*, as it drives demand higher
- An increase in input cost (W) increases P^*, as it drives down the supply.
- The unobserved market heterogeneity $U_S - U_D$ should have a net effect of 0.

The denominator $\alpha_1 - \beta_1$: Measures the difference of sensitivity of demand and supply to price.

- Since $\alpha_1 < 0$ (the demand function has a downward slope) and $\beta_1 > 0$ (the supply function has an upward slope), the denominator is negative

Equilibrium Price Q^*

$$Q^* = \alpha_0 + \alpha_1 \left(\frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D}{\alpha_1 - \beta_1} \right) + \alpha_2 Y + U_D$$

$$Q^* = \frac{\alpha_1 \beta_0 - \beta_1 \alpha_0 + \alpha_1 \beta_2 W - \beta_1 \alpha_2 Y + \alpha_1 U_S - \beta_1 U_D}{\alpha_1 - \beta_1}$$

Interpretation: The equilibrium quantity Q^* has a very similar structure compared to the equilibrium price P^*, except every term in the numerator is weighted by the opposite sensitivity measure. This implies quantity demand is a net effect of both demand and supply factors, weighted by the opposite sensitivity.

Q3 Estimations

3C). Compare your OLS estimates to the true parameters and discuss potential biases. In all tables, put standard errors in parentheses

Table 1: Demand OLS Estimates V.S. True Parameter Values

	True Value	**coef**	**std err**	**t**	**P**$>\|t\|$	**[0.025**	**0.975]**
$\hat{\alpha_0}$	30	22.1133	(0.607)	36.420	0.000	20.922	23.305
$\hat{\alpha_1}$	-2	-0.2910	(0.071)	-4.092	0.000	-0.430	-0.151
$\hat{\alpha_2}$	1	0.6601	(0.053)	12.386	0.000	0.556	0.765

Interpretation: When comparing the point estimates of the parameters for **consumer's quantity demand**, none of them is close to the true parameter value. Even the 95% confidence intervals fail to cover the true parameter values for all point estimates. This suggests the problem of omitted variable bias because we are estimating simultaneous equations where the price P is likely to be correlated with unobserved demand factor U_D, which generates biased point estimates. It also suggests heterogeneity and (or) non-normal errors, which leads to wrong interval estimates.

Table 2: Supply OLS Estimates V.S. True Parameter Values

	True Value	**coef**	**std err**	**t**	**P**$>\|t\|$	**[0.025**	**0.975]**
$\hat{\beta_0}$	5	20.7431	(0.552)	37.607	0.000	19.661	21.825
$\hat{\beta_1}$	3	0.7528	(0.078)	9.659	0.000	0.600	0.906
$\hat{\beta_2}$	2	1.0782	(0.060)	17.986	0.000	0.961	1.196

Interpretation: The supply OLS estimates are even more biased when compared to the true parameters. The concerns are the same as I explained in the above interpretation for the demand OLS estimates

Q4 2SLS Estimates

4a). Select and justify suitable instruments for each equation.

IV for The Demand Equation: W

- **Relevance**:

$$Corr[W, P] \neq 0$$

  - W affects **price (P)** through the **supply equation** (simultaneous equations). Higher input costs reduce supply (Q_S) and vice versa, resulting in a change in the equilibrium price **price (P)**.

- **Exclusion Restriction**:
  - W does *not* directly affect the **demand equation** ($Q_D = \alpha_0 + \alpha_1 P + \alpha_2 Y + U_D$). It only affects demand quantity Q_D indirectly through P.

- **Exogeneity**:

$$Corr(W, U_D) = 0$$

  - W is an exogenous variable that is related to the supply model only, so it is uncorrelated with the demand shock U_D.

Conclusion:

W is a valid instrument for P in the demand equation, because it is relevant ($Corr[W, P] \neq 0$), excluded from the demand equation ($Q_D = \alpha_0 + \alpha_1 P + \alpha_2 Y + U_D$), and exogenous ($Corr(W, U_D) = 0$).

IV for The Supply Equation: Y

- **Relevance**:

$$Corr[Y, P] \neq 0$$

  - Y affects **price (P)** through the **demand equation** (simultaneous equations). Higher input income increases demand (Q_D) and vice versa, resulting in a change in the equilibrium price **price (P)**.

- **Exclusion Restriction**:
  - Y does *not* directly affect the **supply equation** ($Q_S = \beta_0 + \beta_1 P + \beta_2 W + U_S$).It only affects demand quantity Q_S indirectly through P.

- **Exogeneity**:

$$Corr[Y, U_S] = 0$$

  - Y is an exogenous variable that is related to the demand model only, so it is uncorrelated with the demand shock U_S.

Conclusion:

Y is a valid instrument for P in the supply equation, because it is relevant ($Corr[Y, P] \neq 0$), excluded from the demand equation, and exogenous ($Corr(Y, U_S) = 0$).

4c). Compare the OLS and 2SLS results and interpret the differences. In all tables, put standard errors in parentheses.

Table 3: Demand OLS, 2SLS and True Parameter Values

	True Parameter	OLS Estimate	Parameter	Std. Err.	Lower CI	Upper CI
$\hat{\alpha_0}$	30	22.1133	28.600	(0.9179)	26.801	30.399
$\hat{\alpha_1}$	-2	-0.2910	-1.7171	(0.1436)	-1.9986	-1.4356
$\hat{\alpha_2}$	1	0.6601	0.9514	(0.0671)	0.8199	1.0828

Endogenous: P_star
Instruments: W

Table 4: Supply OLS, 2SLS and True Parameter Values

	True Parameter	OLS Estimate	Parameter	Std. Err.	Lower CI	Upper CI
$\hat{\beta}_0$	5	20.7431	4.5548	(2.1700)	0.3017	8.8079
$\hat{\beta}_1$	3	0.7528	3.0520	(0.3083)	2.4478	3.6562
$\hat{\beta}_2$	2	1.0782	2.0921	(0.1509)	1.7963	2.3879

Endogenous: P_star
Instruments: Y

Interpretation All the 2SLS estimates are closer to the true parameter values, and they all successfully include the true parameter values with a 95% confidence interval except for $\hat{\alpha_1}$. By comparing the two sets of estimates, we can determine the signs of bias in the OLS result.

- $\hat{\alpha_0}$: Downward bias
- $\hat{\alpha_1}$: Upward bias
- $\hat{\alpha_2}$: Downward bias
- $\hat{\beta}_0$: Upward bias
- $\hat{\beta}_1$: Downward bias
- $\hat{\beta}_2$: Downward bias

Q5 Per-Unit Tax (Theory)

a) Suppose the government imposes a per-unit tax t. Derive the new equilibrium (P^*_{tax}, Q^*_{tax})

Demand Equation:

$$Q_{D0} = a_0 + a_1 P + a_2 Y + U_D$$

Supply Equation

$$Q_{S0} = \beta_0 + \beta_1 P + \beta_2 W + U_S$$

Demand and Supply with Tax (assuming consumer bears the tax)

New Demand Equation:

$$Q_D^{tax} = a_0 + a_1(P + \omega t) + a_2 Y + U_D$$

New Supply Equation:

$$Q_S^{tax} = \beta_0 + \beta_1(P - (1-\omega)t) + \beta_2 W + U_S$$

where, ω is the share of tax borne by the consumer.

Equilibrium Condition

At equilibrium, $Q_D^{tax} = Q_S^{tax}$

Substituting the demand and supply equations:

$$a_0 + a_1(P + \omega t) + a_2 Y + U_D = \beta_0 + \beta_1(P - (1-\omega)t) + \beta_2 W + U_S$$

Solving for P^*_{tax}:

$$(a_1 - \beta_1)P = \beta_0 - a_0 + \beta_2 W - a_2 Y + U_S - U_D - a_1 \omega t - \beta_1(1-\omega)t(a_1 - \beta_1)$$

$$P = \beta_0 - a_0 + \beta_2 W - a_2 Y + U_S - U_D - t(\omega a_1 + \beta_1 - \omega\beta_1)$$

$$P^*_{tax} = \frac{\beta_0 - a_0 + \beta_2 W - a_2 Y + U_S - U_D - t(\omega a_1 + \beta_1 - \omega\beta_1)}{a_1 - \beta_1}$$

where,
Consumers pay $P^*_{tax} + t$,
Producers receive P^*_{tax}

Solving for Q^*_{tax}:

$$Q^*_{tax} = \beta_0 + \beta_1 \frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D - t(\omega\alpha_1 + \beta_1 - \omega\beta_1)}{\alpha_1 - \beta_1} + \beta_2 W + U_S$$

Simplifying with $\omega = 1$:

$$P^*_{tax} = \frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D - t(\alpha_1 + \beta_1 - \beta_1)}{\alpha_1 - \beta_1} = \frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D - t\alpha_1}{\alpha_1 - \beta_1}$$

$$Q^*_{tax} = \beta_0 + \beta_1 \frac{\beta_0 - \alpha_0 + \beta_2 W - \alpha_2 Y + U_S - U_D - t\alpha_1}{\alpha_1 - \beta_1} + \beta_2 W + U_S$$

b) Write down expressions for consumer surplus, producer surplus, tax revenue, and deadweight loss.

Consumer Surplus

$$Q_D|_{\omega=1} = \alpha_0 + \alpha_1(P + \omega t) + \alpha_2 Y + U_D = \alpha_0 + \alpha_1(P + t) + \alpha_2 Y + U_D$$

$$P_D|_{\omega=1} = \frac{Q_{Dtax} - \alpha_0 - \alpha_2 Y - U_D}{\alpha_1} - \omega t = \frac{Q_{Dtax} - \alpha_0 - \alpha_2 Y - U_D}{\alpha_1} - t$$

$$CS|_{\omega=1} = \int_0^{Q^*_{tax}} \left[\frac{Q_D - \alpha_0 - \alpha_2 Y - U_D}{\alpha_1} - \omega t - P^*_{tax}\right] dQ = \int_0^{Q^*_{tax}} \left[\frac{Q_D - \alpha_0 - \alpha_2 Y - U_D}{\alpha_1} - t - P^*_{tax}\right] dQ$$

Supplier Surplus

$$Q_S|_{\omega=1} = \beta_0 + \beta_1(P - (1-\omega)t) + \beta_2 W + U_S = \beta_0 + \beta_1 P + \beta_2 W + U_S$$

$$P_S|_{\omega=1} = \frac{Q_S - \beta_0 - \beta_2 W - U_S}{\beta_1} + (1-\omega)t = \frac{Q_S - \beta_0 - \beta_2 W - U_S}{\beta_1}$$

$$PS|_{\omega=1} = \int_0^{Q^*_{tax}} \left[\frac{Q_S - \beta_0 - \beta_2 W - U_S}{\beta_1} + (1-\omega)t - P^*_{tax}\right] dQ = \int_0^{Q^*_{tax}} \left[\frac{Q_S - \beta_0 - \beta_2 W - U_S}{\beta_1} - P^*_{tax}\right] dQ$$

Tax Revenue

$$TR = t \cdot Q^*_{tax}$$

Deadweight Loss

$$DWL = \int_{Q^*_{tax}}^{Q^*_0} \left[P^*_{tax} - P_0\right] dQ$$

c). Provide a brief interpretation of how the tax changes market outcomes and welfare.

The Tax reduces the equilibrium quantity by increasing the price paid by the consumer. Now there's a gap between the price paid by the consumers and the price received by the supplier, which goes into the government's pocket as tax revenue. As a result, there is a deadweight loss due to forgone trades that would have taken place. Total welfare reduced.

Q6 Policy Impact (Empirical)

b). For each tax value t, compute consumer surplus, producer surplus, tax revenue, and deadweight loss. Plot these outcomes against t. Interpret how higher taxes affect prices, quantities, and welfare.

- **Consumer & Producer's Surplus:** Figure 1's top panel implies that surpluses to both consumers and suppliers shrink as taxes increase. The forgone surplus partially goes to the government as tax revenue, and the rest is lost due to lost efficiency.
- **Tax Revenue:** Figure 1's bottom left panel implies the diminishing return on tax revenue as tax increases. The initial sharp increase in the tax revenue is due to the fact that a relatively small tax rate is applied to a big stable quantity of transactions. However, as the tax rate continues to increase, it discourages market transactions. Eventually, the decline in the taxable transactions outweighs the revenue gains from the higher tax rate, leading to a net decrease in total revenue.
- **Deadweight Loss:** Figure 1's bottom right panel implies the increasing loss on social welfare as tax increases. As tax spikes, more and more trades are gone which could have taken place, as a result, the decrease in social welfare worsens as the tax goes up.

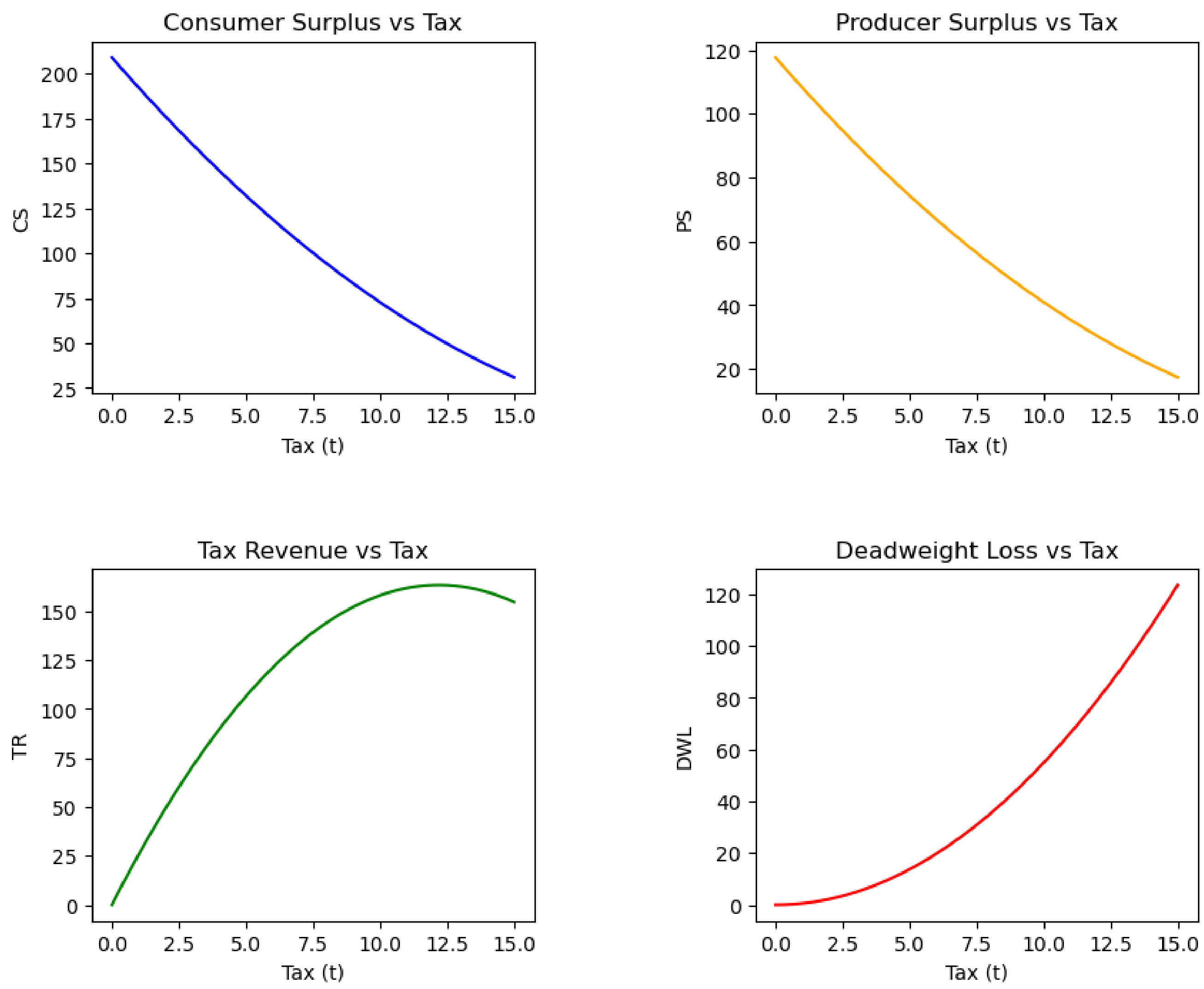

Figure 1: Consumer Surplus, Producer Surplus, Tax Revenue, and Deadweight Loss vs Tax